# Literature Review: River Flow Estimation via Altimetry 06/07/26

## Introduction:

River flow velocities are an essential variable that can influence flood forecasts, erosion, water resources and ecosystem health. Accurate measurements are therefore vital for monitoring rivers and understanding the impact of climate change and weather events. Conventional, in-situ measurements normally require equipment to be used in the river channel. This method can be accurate but also is labour intensive, expensive and unavailable if rivers aren't safely accessible. Non-contact measurement techniques are being developed to overcome these issues. Using radar as a non-contact method is particularly attractive because it can be operated independently of time of day and is unaffected by weather so is a useful continuous measuring system. The most promising developments include Doppler radar and satellite radar altimetry which are still being investigated to estimate river flow velocities at large scales.

## Ground Based Radar Methods:

Ground based non-contact radar systems mostly operate using Doppler shift. The shift in frequency of the transmitted and reflected microwaves is measured and used to infer the velocity of the water surface. Turbulence causes surface roughness which allows the backscatter signals to be detected. (2) The measurements rely on Bragg scattering, where the microwaves are reflected by surface waves of a particular wavelength. (3) This means a certain level of surface roughness is needed for reliable velocity estimation, so rivers with low turbulence can produce weaker signals and lower accuracy even if the velocity is high.

There are many Doppler radar systems that have been investigated with the main three being: fixed-point, side-scan, pulsed and Ultra High Frequency (UHF) systems. Fixed-point systems measure velocity over a small region of the river surface, whereas side-scan systems observe a much larger section of the channel, providing greater spatial coverage. (3) Pulsed Doppler systems can estimate velocity across multiple range bins, whereas Ultra High Frequency (UHF) radar systems use multiple antennas to determine the direction of the returned signal and improve spatial characterisation of river flow. (4)

These methods provide the classic advantages of non-contact methods such as continuous monitoring by operation at any time of day and independent of weather conditions. However, their performance relies on turbulence as rough surfaces are needed to receive appropriate signals. Wind induced surface motion can also introduce errors if not corrected, for example in Alimeti, F. et al, 2020, the wind conditions were chosen as light, but no wind correction applied. Straight channels with uniform steady flow are preferred due to estimations such as Entropy-based Velocity Model (EVM).

# Satellite Altimetry:

Satellite altimetry provides larger spatial coverage than Doppler radar by using observations across regional/ global scales which enables monitoring in remote or inaccessible areas where in-situ is not possible.

Satellite radar altimetry estimates water surface elevation by transmitting microwave pulses towards the Earth's surface and measuring the time taken for the reflected signal to return to the satellite as the pulse interacts with the rough water surface. Since microwaves travel at the speed of light, this travel time can be converted into the distance between the satellite and the water surface. By combining this range measurement with the precisely known position of the satellite, the elevation of the water surface can be determined.

Compared with Doppler radar, satellite altimetry provides much greater spatial coverage, enabling routine observations of rivers on regional and global scales. Unlike Doppler radar systems, which directly estimate surface velocity through frequency shifts, radar altimetry measures changes in water height and therefore provides an indirect indication of river flow. (5) Despite this limitation, the global coverage and freely available nature of satellite observations make radar altimetry attractive for large-scale river monitoring.

# SWOT and Radar Interferometry:

The Surface Water and Ocean Topography (SWOT) mission provides high resolution microwave observations of rivers, lakes and oceans on an almost global scale. Compared to conventional nadir-looking altimeters, SWOT is designed to be slightly side facing to improve the spatial resolution. (6)

The key innovation of the SWOT mission is the Ka-band Radar Interferometer (KaRIn), which uses two antennas to observe the Earth's surface from slightly different viewing angles. KaRIn employs radar interferometry to generate two-dimensional maps of water surface elevation across two 50km wide swaths which massively improves the spatial coverage. However, SWOT's 21 day repeat cycle means it may not capture rapidly evolving events such as floods, and measurements can still be affected by vegetation, dark water and radar noise over narrow or complex river channels. By comparing the phase difference between the signals received at each antenna, SWOT can retrieve high-resolution water surface elevations for rivers, lakes and floodplains, significantly increasing spatial coverage compared with previous satellite altimeters. (7) Although the primary purpose of the interferometric phase measurements is to derive water surface elevation, the phase observations may also contain information relating to surface motion, motivating investigation into their potential for estimating river flow velocity.

Radar interferometry uses the phase difference between microwaves received at two spatially separated antennas observing the same target. Because the radar signals travel slightly different distances before being received, the resulting phase difference is directly related to the geometry of the reflecting surface. This enables the reconstruction of high-resolution water surface elevations across the swath. Compared with conventional radar altimeters, interferometry substantially improves spatial resolution and enables two-dimensional mapping of inland water bodies rather than measurements along a single nadir track. (7)

The combination of interferometric imaging and high spatial resolution represents a significant advance for inland water monitoring. Previous satellite altimeters were largely limited to nadir observations, restricting their ability to monitor narrow rivers. SWOT overcomes these limitations by producing two-dimensional maps of water surface elevation, width and slope over large scales, enabling better monitoring of river systems. This has considerable potential for improving flood forecasting, hydrological modelling and water resources, particularly in remote regions. (6)

# River Velocity Estimation from SWOT:

Since its launch, SWOT has provided observations of inland water systems through its Level-2 products, including measurements of water surface elevation, river width and water surface slope. However, these products were primarily designed to retrieve water surface topography rather than directly measure river flow velocity. (6)

Although water surface elevation, slope and river width provide valuable insight into river behaviour, they do not directly describe the motion of the water itself. River flow velocity is influenced by a range of factors, including channel morphology, meaning that relationships between these variables and velocity may not always be straightforward. As a result, while existing SWOT products may contain information related to river flow, additional observations or alternative approaches may be required to estimate surface velocity.

Existing work has largely focused on products derived from SWOT, while little work has been done on whether the underlying interferometric observations contain information related to river motion. Since Doppler radar exploits phase shifts due to moving water, investigating similar signatures within SWOT observations represents a promising direction. Recent work by Thurman, H.R. *et al.* (2025) used SWOT to find direct observations of downstream changes in river height within a flow wave and produced spatial hydrographs as in-situ analogues. (7) Other papers such as Lichtman, I.D. *et al.* (2025) also investigate the validity of using SWOT data to estimate river height and compare against in-situ measurements to prove the accuracy is to a high degree. For example, in the Bristol Channel-Severn Estuary, the scaled L3 data exhibit an RMSD of 0.137 m, a regression slope of 0.99 and offset +0.044 m,

Beyond the processed Level-2 products, SWOT also acquires Level-1A observations containing the original radar measurements from the Ka-band Radar Interferometer. These raw data preserve the complex radar signal, including the interferometric phase information used during later processing to derive water surface elevation. Higher-level products are generated through multiple processing stages so analysing the original interferometric observations provides access to information that may not be fully retained at later levels. (8)

Interferometric phase is sensitive to changes in the propagation path of the radar signal and has been widely exploited within remote sensing to retrieve information about surface geometry. (9) While previous studies have focused predominantly on the retrieval of water surface elevation from SWOT observations, little attention has been given to the potential for estimating river surface velocity. This project therefore investigates whether river velocity can first be inferred from existing SWOT products and, where these prove insufficient, whether additional information contained within the raw interferometric phase observations may improve velocity estimation. Establishing this would be important for satellite-based monitoring of river velocity on a global scale.

# Conclusion:

Ground-based Doppler radar and satellite radar altimetry each provide advantages for river monitoring, with Doppler systems directly measuring surface velocity and satellite observations offering high spatial coverage. The development of SWOT has substantially advanced satellite hydrology through interferometric observations capable of measuring river height, width and slope at high spatial resolution. Despite these advances, direct estimation of river flow velocity from satellite observations remains largely unexplored. This literature therefore motivates the present study, which investigates whether river velocity can first be inferred from existing SWOT-derived products before assessing whether additional information contained within Level-1A interferometric phase observations may improve velocity estimation.

# References:


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